American Computer Science League

2023-2024 ● Contest 3: Short Problems ● Intermediate Division

1. Boolean Algebra

How many ordered triples make this expression TRUE?

$$A(\overline{\overline{B} + C})(\overline{AB + BC})$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. Data Structures

Find the number of nodes that have only 1 child in the binary search tree for:

PONTIACVIBE

A. 4
B. 5
C. 6
D. 7
E. 8

3. Data Structures

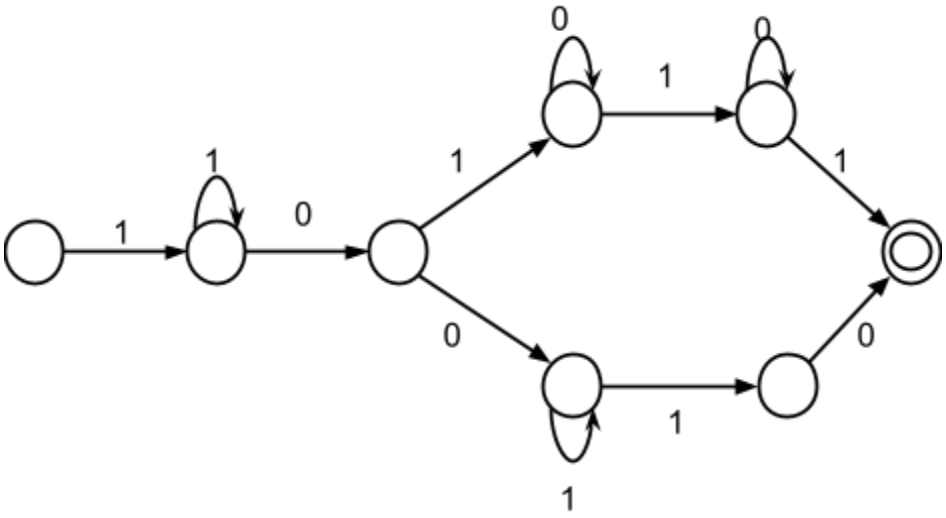
What does the final stack look like from the top to the bottom after performing the following algorithm on an initially empty stack when using the word "ALPHABETS"?

```
for letter = each letter in a word
    if the letter is a vowel
        PUSH(letter)
    else
        x = POP()
        PUSH(letter)
        PUSH(x)
    end if
next letter
```

A. ALPHABETS
B. ESTABAHPL
C. LPHABATSE
D. HPLABASTE
E. STEBAHPLA

4. FSAs

How many of the following strings satisfy the FSA below?



a. 1101011010 b. 11010 c. 1111111

d. 1010101 e. 1100110 f. 1001110

A. 1
B. 2
C. 3
D. 4
E. 5

5. Regular Expressions

Today the pop-up ACSL Coffee Shop is open for business. Only coffee that satisfies the following regular expression may be ordered. How many different kinds of coffee are available?

[b-j] [aeiou] [^k]*

regular	decaf	latte
dark	drip	iced
espresso	black	cappuccino

A. 2
B. 3
C. 4
D. 5
E. 6

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1. 布尔代数

请问**有多少个有序三元组可以使得以下表达式的结果为真？**

$$\overline{A(\overline{B} + C)}\,\overline{(AB + BC)}$$

A. 0
B. 1
C. 2
D. 3
E. 4

2. 数据结构

请问**在以下字符串**构成**的二叉搜索树中，有多少个节点只有一个子节点？**

PONTIACVIBE

A. 4
B. 5
C. 6
D. 7
E. 8

3. 数据结构

使用**字母**“ALPHABETS”，**在一个初始为空的栈中执行下列算法，求最终从上到下出栈的结果。**

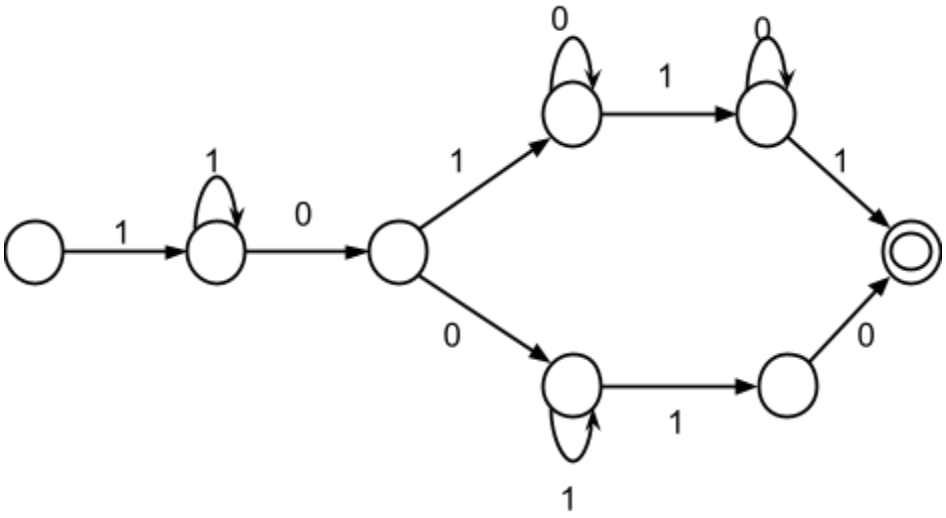
```
for letter = each letter in a word
    if the letter is a vowel
        PUSH(letter)
    else
        x = POP()
        PUSH(letter)
        PUSH(x)
    end if
next letter
```

（注：vowel 指元音字母）

A. ALPHABETS
B. ESTABAHPL
C. LPHABATSE
D. HPLABASTE
E. STEBAHPLA

4. 状态转换图

请问**有多少个字符串满足以**下状态转换图？



a. 1101011010　　b. 11010　　c. 1111111

d. 1010101　　e. 1100110　　f. 1001110

A. 1
B. 2
C. 3
D. 4
E. 5

5. 正则表达式

ACSL快闪**咖啡店今日**开**业。只能订购满足下列正则表达式的咖啡。请问有多少种不同咖啡**可供订购**？**

[b-j] [aeiou] [^k]*

regular	decaf	latte
dark	drip	iced
espresso	black	cappuccino

A. 2
B. 3
C. 4
D. 5
E. 6